KG4Ex-NOV-Set on Eedi | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.964 ± 0.006	0.956 ± 0.007	**0.980 ± 0.000**	0.971 ± 0.001	0.950 ± 0.010	0.961 ± 0.004
10	0.955 ± 0.009	0.939 ± 0.004	**0.979 ± 0.000**	0.959 ± 0.002	0.938 ± 0.009	0.944 ± 0.005
15	0.938 ± 0.016	0.921 ± 0.014	**0.969 ± 0.006**	0.947 ± 0.002	0.927 ± 0.004	0.927 ± 0.006
20	0.924 ± 0.008	0.898 ± 0.014	**0.958 ± 0.005**	0.934 ± 0.003	0.919 ± 0.003	0.912 ± 0.006
25	0.916 ± 0.010	0.881 ± 0.010	**0.942 ± 0.010**	0.923 ± 0.003	0.912 ± 0.005	0.897 ± 0.007
30	0.910 ± 0.008	0.869 ± 0.007	0.914 ± 0.006	**0.914 ± 0.002**	0.904 ± 0.004	0.883 ± 0.007
35	**0.906 ± 0.008**	0.861 ± 0.006	0.900 ± 0.008	0.904 ± 0.001	0.897 ± 0.005	0.870 ± 0.006
40	**0.903 ± 0.008**	0.854 ± 0.007	0.887 ± 0.010	0.894 ± 0.002	0.886 ± 0.004	0.859 ± 0.006
45	**0.897 ± 0.008**	0.846 ± 0.010	0.874 ± 0.006	0.884 ± 0.002	0.878 ± 0.002	0.849 ± 0.005
50	**0.890 ± 0.003**	0.837 ± 0.017	0.869 ± 0.005	0.877 ± 0.001	0.870 ± 0.005	0.840 ± 0.004

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

C(u) is the student's correct-response concept set. Average students first, then report mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV-Set on Eedi | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	0.964 ± 0.006	0.862	0.890	**0.971**	0.971 ± 0.001	0.966 ± 0.002	0.969 ± 0.000	0.968 ± 0.002	0.965 ± 0.002
10	0.955 ± 0.009	0.759	0.810	**0.964**	0.958 ± 0.001	0.956 ± 0.003	0.958 ± 0.001	0.959 ± 0.002	0.958 ± 0.004
15	0.938 ± 0.016	0.676	0.752	**0.959**	0.948 ± 0.001	0.947 ± 0.004	0.949 ± 0.001	0.952 ± 0.004	0.952 ± 0.002
20	0.924 ± 0.008	0.611	0.705	**0.950**	0.936 ± 0.002	0.935 ± 0.003	0.940 ± 0.002	0.942 ± 0.003	0.944 ± 0.004
25	0.916 ± 0.010	0.563	0.666	**0.943**	0.926 ± 0.002	0.925 ± 0.004	0.932 ± 0.001	0.932 ± 0.003	0.935 ± 0.003
30	0.910 ± 0.008	0.525	0.632	**0.935**	0.918 ± 0.002	0.915 ± 0.004	0.923 ± 0.001	0.925 ± 0.003	0.928 ± 0.002
35	0.906 ± 0.008	0.495	0.602	**0.929**	0.908 ± 0.001	0.906 ± 0.005	0.916 ± 0.001	0.916 ± 0.002	0.921 ± 0.002
40	0.903 ± 0.008	0.469	0.575	**0.923**	0.899 ± 0.001	0.896 ± 0.005	0.910 ± 0.002	0.906 ± 0.002	0.914 ± 0.002
45	0.897 ± 0.008	0.453	0.551	**0.917**	0.890 ± 0.001	0.888 ± 0.006	0.904 ± 0.002	0.900 ± 0.001	0.907 ± 0.002
50	0.890 ± 0.003	0.434	0.537	**0.909**	0.881 ± 0.001	0.879 ± 0.006	0.898 ± 0.001	0.893 ± 0.002	0.900 ± 0.002

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

C(u) is the student's correct-response concept set. Average students first, then report mean ± sample SD across seeds 2024, 2025, 2026.

EB-CF, SB-CF and CBF are deterministic single runs. Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV-Set on Algebra2005 | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.984 ± 0.002	0.865 ± 0.018	**0.999 ± 0.000**	0.975 ± 0.005	0.976 ± 0.003	0.969 ± 0.002
10	0.958 ± 0.009	0.840 ± 0.029	**0.999 ± 0.000**	0.954 ± 0.009	0.952 ± 0.005	0.945 ± 0.002
15	0.943 ± 0.008	0.827 ± 0.034	**0.999 ± 0.000**	0.936 ± 0.013	0.938 ± 0.008	0.923 ± 0.004
20	0.941 ± 0.009	0.804 ± 0.045	**0.985 ± 0.024**	0.914 ± 0.011	0.930 ± 0.005	0.900 ± 0.006
25	0.931 ± 0.009	0.785 ± 0.057	**0.967 ± 0.025**	0.895 ± 0.009	0.922 ± 0.002	0.881 ± 0.009
30	0.916 ± 0.003	0.760 ± 0.051	**0.947 ± 0.009**	0.876 ± 0.011	0.910 ± 0.002	0.859 ± 0.007
35	0.904 ± 0.009	0.734 ± 0.039	**0.942 ± 0.000**	0.857 ± 0.014	0.900 ± 0.004	0.838 ± 0.009
40	0.894 ± 0.012	0.700 ± 0.017	**0.942 ± 0.000**	0.838 ± 0.015	0.892 ± 0.004	0.817 ± 0.008
45	0.879 ± 0.014	0.678 ± 0.011	**0.936 ± 0.009**	0.816 ± 0.014	0.884 ± 0.009	0.794 ± 0.011
50	0.864 ± 0.004	0.661 ± 0.009	**0.931 ± 0.018**	0.795 ± 0.010	0.872 ± 0.009	0.773 ± 0.007

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

C(u) is the student's correct-response concept set. Average students first, then report mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV-Set on Algebra2005 | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	**0.984 ± 0.002**	0.541	0.750	0.927	0.970 ± 0.002	0.975 ± 0.002	0.977 ± 0.002	0.890 ± 0.004	0.771 ± 0.007
10	**0.958 ± 0.009**	0.474	0.709	0.895	0.949 ± 0.003	0.957 ± 0.003	0.936 ± 0.006	0.844 ± 0.009	0.683 ± 0.009
15	0.943 ± 0.008	0.441	0.645	0.857	0.929 ± 0.005	**0.945 ± 0.003**	0.917 ± 0.006	0.806 ± 0.011	0.625 ± 0.007
20	**0.941 ± 0.009**	0.414	0.598	0.829	0.903 ± 0.011	0.928 ± 0.010	0.888 ± 0.006	0.779 ± 0.011	0.585 ± 0.014
25	**0.931 ± 0.009**	0.382	0.559	0.804	0.883 ± 0.012	0.914 ± 0.012	0.872 ± 0.007	0.746 ± 0.009	0.547 ± 0.008
30	**0.916 ± 0.003**	0.364	0.531	0.778	0.866 ± 0.012	0.901 ± 0.011	0.855 ± 0.007	0.720 ± 0.013	0.520 ± 0.007
35	**0.904 ± 0.009**	0.351	0.510	0.763	0.851 ± 0.010	0.884 ± 0.010	0.836 ± 0.008	0.697 ± 0.013	0.501 ± 0.007
40	**0.894 ± 0.012**	0.340	0.498	0.745	0.834 ± 0.007	0.871 ± 0.014	0.818 ± 0.005	0.675 ± 0.009	0.485 ± 0.010
45	**0.879 ± 0.014**	0.334	0.497	0.722	0.819 ± 0.006	0.852 ± 0.017	0.794 ± 0.006	0.656 ± 0.007	0.473 ± 0.010
50	**0.864 ± 0.004**	0.332	0.498	0.706	0.804 ± 0.008	0.832 ± 0.014	0.771 ± 0.006	0.637 ± 0.008	0.462 ± 0.010

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

C(u) is the student's correct-response concept set. Average students first, then report mean ± sample SD across seeds 2024, 2025, 2026.

EB-CF, SB-CF and CBF are deterministic single runs. Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV-Set on XES3G5M-sub-small | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.988 ± 0.001	**0.994 ± 0.003**	0.991 ± 0.002	0.990 ± 0.000	0.985 ± 0.001	0.988 ± 0.001
10	0.981 ± 0.002	0.982 ± 0.004	0.982 ± 0.005	**0.984 ± 0.001**	0.975 ± 0.003	0.980 ± 0.001
15	0.975 ± 0.002	0.977 ± 0.004	0.974 ± 0.007	**0.978 ± 0.001**	0.966 ± 0.003	0.974 ± 0.001
20	0.968 ± 0.004	**0.973 ± 0.004**	0.970 ± 0.008	0.972 ± 0.001	0.960 ± 0.003	0.967 ± 0.001
25	0.962 ± 0.006	**0.971 ± 0.004**	0.963 ± 0.008	0.966 ± 0.001	0.955 ± 0.004	0.961 ± 0.001
30	0.957 ± 0.006	**0.967 ± 0.002**	0.956 ± 0.008	0.960 ± 0.001	0.950 ± 0.006	0.955 ± 0.001
35	0.954 ± 0.007	**0.965 ± 0.002**	0.950 ± 0.009	0.954 ± 0.001	0.946 ± 0.007	0.949 ± 0.001
40	0.949 ± 0.005	**0.962 ± 0.003**	0.945 ± 0.008	0.949 ± 0.001	0.941 ± 0.007	0.944 ± 0.001
45	0.944 ± 0.004	**0.960 ± 0.003**	0.939 ± 0.008	0.944 ± 0.001	0.936 ± 0.007	0.939 ± 0.001
50	0.939 ± 0.003	**0.954 ± 0.002**	0.935 ± 0.008	0.939 ± 0.001	0.932 ± 0.007	0.934 ± 0.001

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

C(u) is the student's correct-response concept set. Average students first, then report mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV-Set on XES3G5M-sub-small | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Union of concepts in each student's Top-N list; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	0.988 ± 0.001	0.959	0.971	0.984	0.988 ± 0.001	0.988 ± 0.000	0.969 ± 0.001	**0.991 ± 0.001**	0.989 ± 0.000
10	0.981 ± 0.002	0.929	0.949	0.975	0.983 ± 0.000	0.983 ± 0.000	0.957 ± 0.001	**0.987 ± 0.000**	0.983 ± 0.001
15	0.975 ± 0.002	0.900	0.929	0.967	0.979 ± 0.001	0.978 ± 0.000	0.949 ± 0.000	**0.983 ± 0.000**	0.978 ± 0.000
20	0.968 ± 0.004	0.875	0.913	0.961	0.976 ± 0.001	0.973 ± 0.000	0.942 ± 0.001	**0.979 ± 0.000**	0.972 ± 0.001
25	0.962 ± 0.006	0.849	0.894	0.954	0.969 ± 0.000	0.965 ± 0.000	0.934 ± 0.001	**0.975 ± 0.000**	0.967 ± 0.001
30	0.957 ± 0.006	0.824	0.876	0.947	0.963 ± 0.000	0.958 ± 0.001	0.928 ± 0.001	**0.971 ± 0.001**	0.961 ± 0.001
35	0.954 ± 0.007	0.801	0.856	0.941	0.958 ± 0.001	0.952 ± 0.001	0.922 ± 0.001	**0.967 ± 0.001**	0.956 ± 0.001
40	0.949 ± 0.005	0.780	0.837	0.935	0.953 ± 0.001	0.945 ± 0.001	0.915 ± 0.000	**0.963 ± 0.001**	0.951 ± 0.001
45	0.944 ± 0.004	0.761	0.820	0.927	0.948 ± 0.001	0.938 ± 0.002	0.908 ± 0.001	**0.960 ± 0.001**	0.946 ± 0.000
50	0.939 ± 0.003	0.741	0.804	0.921	0.943 ± 0.001	0.931 ± 0.002	0.903 ± 0.000	**0.956 ± 0.001**	0.941 ± 0.001

K_TopN(u) = union of concepts across the Top-N exercises; NOV-Set(u) = 1 - |K_TopN(u) intersection C(u)| / |K_TopN(u) union C(u)|.

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